

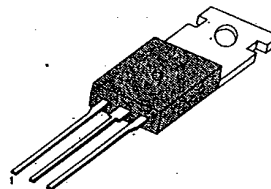
TIP42 SERIES**(TIP42/42A/42B/42C) PNP EXITAXIAL SILICON TRANSISTOR****MEDIUM POWER LINEAR
SWITCHING APPLICATIONS**

- Complement to TIP41/41A/41B/41C

ABSOLUTE MAXIMUM RATINGS ($T_a=25^\circ\text{C}$)

Characteristic	Symbol	Rating	Unit
Collector-Base Voltage : TIP42	V_{CB0}	-40	V
: TIP42A		-60	V
: TIP42B		-80	V
: TIP42C		-100	V
Collector-Emitter Voltage : TIP42	V_{CE0}	-40	V
: TIP42A		-60	V
: TIP42B		-80	V
: TIP42C		-100	V
Emitter-Base Voltage	V_{EB0}	-5	V
Collector Current (DC)	I_C	-6	A
Collector Current (Pulse)	I_C	-10	A
Base Current	I_B	-2	A
Collector Dissipation ($T_c=25^\circ\text{C}$)	P_C	65	W
Collector Dissipation ($T_a=25^\circ\text{C}$)	P_C	2	W
Junction Temperature	T_j	150	$^\circ\text{C}$
Storage Temperature	T_{stg}	-65~150	$^\circ\text{C}$

TO-220



1. Base 2. Collector 3. Emitter

3

ELECTRICAL CHARACTERISTICS ($T_c=25^\circ\text{C}$)

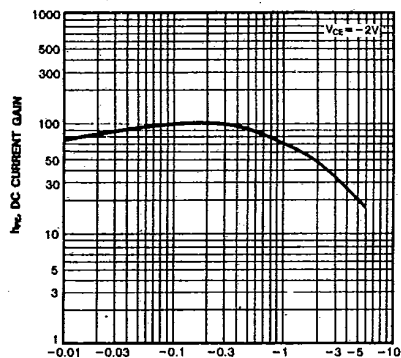
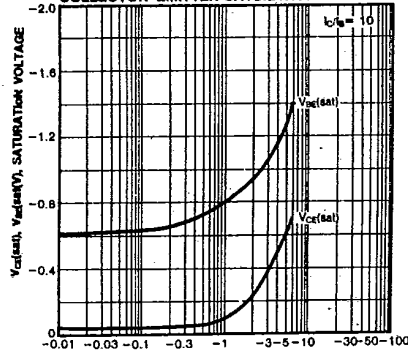
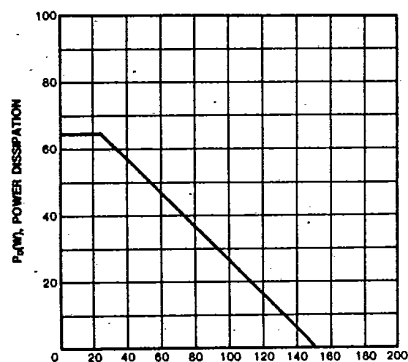
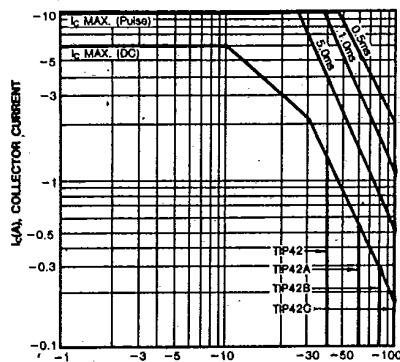
Characteristic	Symbol	Test Condition	Min	Max	Unit
* Collector Emitter Sustaining Voltage : TIP42	$BV_{CE0}(\text{sus})$	$I_C = -30\text{mA}, I_B = 0$	-40		V
: TIP42A			-60		V
: TIP42B			-80		V
: TIP42C			-100		V
Collector Cutoff Current : TIP42/42A	I_{CE0}	$V_{CE} = -30\text{V}, I_B = 0$		-0.7	mA
: TIP42B/42C		$V_{CE} = -60\text{V}, I_B = 0$		-0.7	mA
Collector Cutoff Current : TIP42	I_{CES}	$V_{CE} = -40\text{V}, V_{EB} = 0$		-400	μA
: TIP42A		$V_{CE} = -60\text{V}, V_{EB} = 0$		-400	μA
: TIP42B		$V_{CE} = -80\text{V}, V_{EB} = 0$		-400	μA
: TIP42C		$V_{CE} = -100\text{V}, V_{EB} = 0$		-400	μA
Emitter Cutoff Current	I_{EBO}	$V_{BE} = -5\text{V}, I_C = 0$		-1	mA
* DC Current Gain	h_{FE}	$V_{CE} = -4\text{V}, I_C = -0.3\text{A}$	30		
		$V_{CE} = -4\text{V}, I_C = -3\text{A}$	15	75	
* Collector-Emitter Saturation Voltage	$V_{CE}(\text{sat})$	$I_C = -6\text{A}, I_B = -600\text{mA}$		-1.5	V
* Base-Emitter On Voltage	$V_{BE}(\text{on})$	$V_{CE} = -4\text{V}, I_C = -6\text{A}$		-2.0	V
Current Gain Bandwidth Product	f_T	$V_{CE} = -10\text{V}, I_C = -500\text{mA}$	3.0		MHz
		$f = 1\text{MHz}$			

- * Pulse Test: $PW \leq 300\mu\text{s}$, Duty Cycle $\leq 2\%$



TIP42 SERIES**(TIP42/42A/42B/42C) PNP EXITAXIAL SILICON TRANSISTOR**

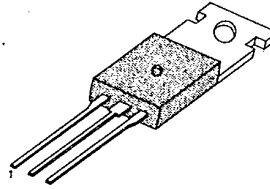
T-33-21

DC CURRENT GAIN I_C (A), COLLECTOR CURRENT**BASE-EMITTER SATURATION VOLTAGE
COLLECTOR-EMITTER SATURATION VOLTAGE** I_C (A), COLLECTOR CURRENT**POWER DERATING** T_d (°C), CASE TEMPERATURE**SAFE OPERATING AREA** V_{CE} (V), COLLECTOR-EMITTER VOLTAGE

TIP47/48/49/50**NPN SILICON TRANSISTOR***T-33-11***HIGH VOLTAGE AND SWITCHING APPLICATIONS****HIGH SUSTAINING VOLTAGE****(V_{ce(sus)}: 250 to 400V)****1A RETED COLLECTOR CURRENT****ABSOLUTE MAXIMUM RATINGS (T_a=25°C)**

Characteristic	Symbol	Rating	Unit
Collector-Base Voltage : TIP47	V _{CB0}	350	V
: TIP48		400	V
: TIP49		450	V
: TIP50		500	V
Collector-Emitter Voltage : TIP47	V _{CEO}	250	V
: TIP48		300	V
: TIP49		350	V
: TIP50		400	V
Emitter-Base Voltage	V _{EB0}	5	V
Collector Current (DC)	I _C	1	A
Collector Current (Pulse)	I _C	2	A
Base Current	I _B	0.6	A
Collector Dissipation (T _C =25°C)	P _C	40	W
Collector Dissipation (T _a =25°C)	P _C	2	W
Junction Temperature	T _J	150	°C
Storage Temperature	T _{stg}	-65~150	°C

TO-220



1. Base 2. Collector 3. Emitter

3**ELECTRICAL CHARACTERISTICS (T_a=25°C)**

Characteristic	Symbol	Test Condition	Min	Max	Unit
Collector Emitter Sustaining Voltage	V _{CEX(SUS)}	I _C =30mA, I _B =0	250 300 350 400		V
: TIP47					
: TIP48					
: TIP49					
: TIP50					
Collector Cutoff Current	I _{CEO}	V _{CE} =150V, I _B =0		1	mA
: TIP47		V _{CE} =200V, I _B =0		1	mA
: TIP48		V _{CE} =250V, I _B =0		1	mA
: TIP49		V _{CE} =300V, I _B =0		1	mA
: TIP50		V _{CE} =350V, V _{BE} =0		1	mA
Collector Cutoff Current	I _{CES}	V _{CE} =400, V _{BE} =0		1	mA
: TIP47		V _{CE} =450V, V _{BE} =0		1	mA
: TIP48		V _{CE} =500V, V _{BE} =0		1	mA
: TIP49		V _{EB} =5V, I _C =0		1	mA
: TIP50				1	mA
Emitter Cutoff Current	I _{EBO}			1	mA
• DC Current Gain	h _{FE}	V _{CE} =10V, I _C =0.3A	30	150	
		V _{CE} =10V, I _C =1A	10		
• Collector Emitter Saturation Voltage	V _{CE(sat)}	I _C =1A, I _B =0.2A		1	V
• Base Emitter On Voltage	V _{BE(on)}	V _{CE} =10V, I _C =1A		1.5	V
Current Gain Bandwidth Product	f _T	V _{CE} =10V, I _C =0.2A, f=1KHz	10		MHz
Turn On Time	t _{on}	V _{CC} =400V		0.5	μs
Storage Time	t _s	5I _{B1} = -2.5I _{B2} , I _C =6A		3	μs
Fall Time	t _f	RL=66.7Ω		0.3	μs

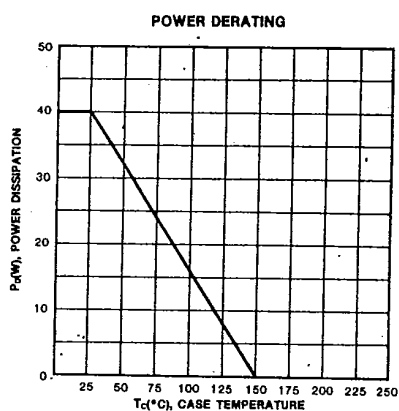
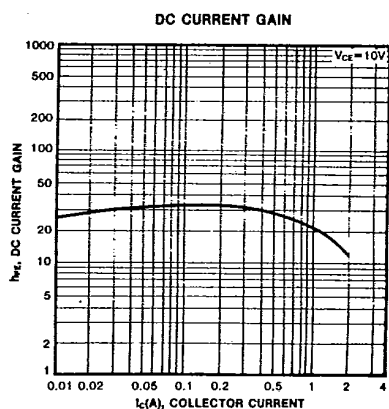
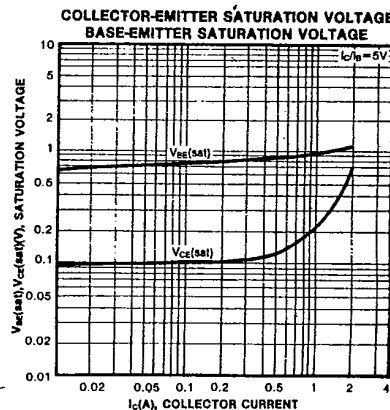
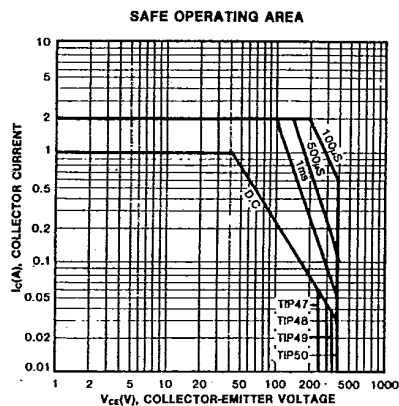
* Pulse test: PW≤300μs, duty cycle ≤ 2% Pulse



TIP47/48/49/50

NPN SILICON TRANSISTOR

T-33-11



TIP100/101/102

SILICON DARLINGTON TRANSISTOR

T-33-29

HIGH DC CURRENT GAIN

MIN $h_{FE}=1000$ @ $V_{CE}=4V$, $I_C=3A$

COLLECTOR-EMITTER SUSTAINING VOLTAGE

LOW COLLECTOR-EMITTER

SATURATION VOLTAGE

MONOLITHIC CONSTRUCTION WITH BUILT

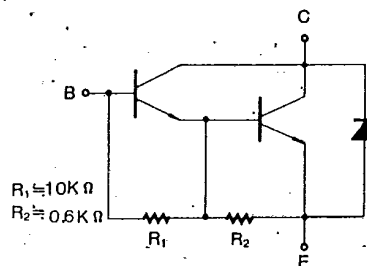
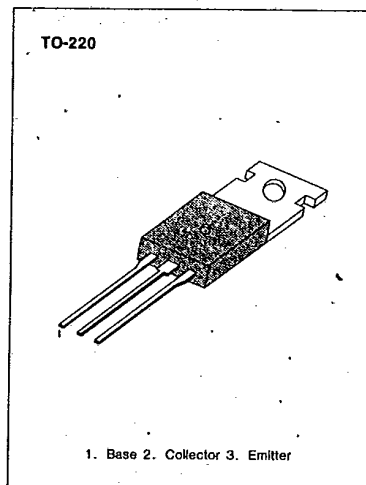
IN BASE-EMITTER SHUNT RESISTORS

INDUSTRIAL USE

Complementary to TIP105/106/107

ABSOLUTE MAXIMUM RATINGS ($T_a=25^\circ C$)

Characteristic	Symbol	Rating	Unit
Collector-Base Voltage : TIP100	V_{CBO}	60	V
: TIP101		80	V
: TIP102		100	V
Collector-Emitter Voltage	V_{CEO}	60	V
: TIP100		80	V
: TIP101		100	V
Emitter-Base Voltage	V_{EBO}	5	V
Collector Current (DC)	I_C	8	A
Collector Current (Pulse)	I_C	15	A
Base Current (DC)	I_B	1	A
Collector Dissipation ($T_a=25^\circ C$)	P_C	2	W
Collector Dissipation ($T_c=25^\circ C$)	P_C	80	W
Junction Temperature	T_J	150	$^\circ C$
Storage Temperature	T_{stg}	-65~150	$^\circ C$

ELECTRICAL CHARACTERISTICS ($T_a=25^\circ C$)

Characteristic	Symbol	Test Condition	Min	Max	Unit
Collector Emitter Sustaining Voltage	$V_{CE(sus)}$	$I_C=30mA$, $I_B=0$	60 80 100		V
: TIP100					
: TIP101					
Collector Cutoff Current : TIP100	I_{CEO}	$V_{CE}=30V$, $I_B=0$		50	μA
: TIP101		$V_{CE}=40V$, $I_B=0$		50	μA
: TIP102		$V_{CE}=50V$, $I_B=0$		50	μA
Collector Cutoff Current : TIP100	I_{CBO}	$V_{CB}=60V$, $I_E=0$		50	μA
: TIP101		$V_{CB}=80V$, $I_E=0$		50	μA
: TIP102		$V_{CB}=100V$, $I_E=0$		50	μA
Emitter Cutoff Current	I_{EBO}	$V_{BE}=5V$, $I_C=0$		2	mA
DC Current Gain	h_{FE}	$V_{CE}=4V$, $I_C=3A$	1000 200	20000	
		$V_{CE}=4V$, $I_C=8A$			
Collector Emitter Saturation Voltage	$V_{CE(sat)}$	$I_C=3A$, $I_B=6mA$		2	V
		$I_C=8A$, $I_B=80mA$		2.5	V
Base Emitter On Voltage	$V_{BE(on)}$	$V_{CE}=4V$, $I_C=8A$		2.8	V
Output Capacitance	C_{ob}	$V_{CB}=10V$, $I_E=0$, $f=0.1MHz$		200	pF

